Curriculum Vitae

Byron Jacobs

1 Personal Details

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Date of Birth: 10 June 1987

Research Profile

NRF Rating:	Y2 (01 January 2020 to 31 December 2025.)
Number of Publication:	43 (Google Scholar)
Research Interests:	Applied Machine Learning, Biomedical Image Processing, Fractional Differential Equations, Numerical Methods, Numerical Analysis.

Citations	All	Since 2021
Citations	404	280
h-index	14	12
i10-index	18	15

2 Qualifications

2.1 Tertiary Education

Year	Qualification
2006 - 2009	BSc (Wits) Computational and Applied Mathematics
2010	BSc Hons with Distinction (Wits) Computational and Applied Mathematics
2011 - 2012	MSc (Wits) Computational and Applied Mathematics
2012 - 2013	Conversion to PhD (Wits) in Computational and Applied Mathematics
2011 - 2014	PhD submitted in February 2014 and Awarded in May 2014.

MSc title: *Noise Reduction Using the Fitzhugh-Nagumo Equation and The Effects of Partial Differential Equations Applied to Image Processing*
PhD title: *On the Application of Partial Differential Equations and Fractional Partial Differential Equations to Images and Their Methods of Solution*

3 Academic Citizenship

3.1 Positions Held

2010	Tutor to undergraduate courses.
2011 - 2012	Appointed as sessional lecturer in the School of Computational and Applied Mathematics, University of the Witwatersrand.
2012 - 2014	Appointed as an Associate Lecturer.
2014 - 2016	Appointed as lecturer in the School of Computational and Applied Mathematics.
2016 - 2018	Appointed as an adjunct lecturer in the School of Mathematics and Statistics at the University of New South Wales.
2016 - 2018	Appointed as senior lecturer in the School of Computer Science and Applied Mathematics, University of the Witwatersrand.
2019 - 2020	Appointed as Associate Professor in the School of Computer Science and Applied Mathematics, University of the Witwatersrand.
2020 - 2024	Appointed as Associate Professor in the Department of Mathematics and Applied Mathematics, University of Johannesburg.
2024 -	Appointed as Visiting Researcher in the Department of Mathematics and Applied Mathematics, University of Johannesburg.
2020 -	Co-founder and Chief Science Officer of VitruvianMD.

3.2 Departmental Administration

2012 - 2014	Open Day representative for the School of Computational and Applied Mathematics.
2012 - 2015	Second year course coordinator.
2012 - 2020	Assisting with Registration (School and Faculty desk)
2013 - 2014	Acting Honours project coordinator.
2013 - 2020	Member of the school Teaching and Learning Committee.
2014 - 2016	Project Leader for the Postgraduate Mentoring Programme (PGMP).
2016 - 2020	Mentor within the Postgraduate Mentoring Programme (PGMP).
2014 - 2020	Occupational Health, Safety and Environment Representative for the School of Computer Science and Applied Mathematics.
2015 - 2020	Industry liaison with Standard Bank, Nedbank, Discovery, Kopano. (organising talks, preinterview testing, industry engagement days (innovation day))
2015 - 2016	Honours course coordinator.
2016 - 2020	Postgraduate coordinator.
2016 - 2020	Chairperson of the school Postgraduate committee.
2017 - 2020	Member of the School executive.
2020 -	Industry liaison, graduate employ-ability organisation.

3.3 University Administration

2014 -	Member of the Wits Readmissions Committee.
2016 -	Graduate Studies Committee Representative for CSAM.
2016 -	Chairperson of the school Occupational Health, Safety and Environment.

3.4 Conference Organisation

2014	Chairperson of the organising committee for the 20 Years of Democracy Symposium for Mathematics in South Africa .
2013 - 2014	Member of the local organising committee for the annual South African Symposium on Numerical and Applied Mathematics (SANUM).
2015	Member of the local organising committee for the annual South African Mathematics Society (SAMS) Congress.
2016 - 2017	Chairperson of the organising committee for the national/international South African Symposium on Numerical and Applied Mathematics (SANUM).
2022 - 2023	Chairperson of the organising committee for the national/international South African Symposium on Numerical and Applied Mathematics (SANUM) for the first time at the University of Johannesburg.

3.5 Workshop Participation

2010	Competed at the International Genetically Engineered Machine Competition hosted by The Massachusetts Institute of Technology.
2011 - 2015	Mathematics in Industry Study Group.
2011	Center for High Performance Computing Winter School.
2011	Attended the STLC 'Professionalisation of Teaching and Learning in Science' course.
2011	Presented work at the 'Professionalisation of Teaching' Symposium.
2012	Oxford Centre for Collaborative Applied Mathematics 4th Graduate Modelling Camp.
2012	85th European Study Group with Industry.
2012	Writing workshop held at Pullen Farm facilitated with Prof Kevin Balkwill
2013	Induction Programme at the Science Teaching and Learning Centre (STLC).
2013	Attended and presented at the Science Teaching and Learning Symposium.
2014	PG Supervision Core course at Centre for Learning Teaching and Development (CLTD).
2015	Planning and academic career course at CLTD.
2015	Publishing for impact course at CLTD.
2015	Writing winning proposals course at CLTD.
2016	Successful completion of Coursera Machine Learning - Stanford University.
2017	Presentation on Sound Synthesis for the community engagement workshop hosted by the Academy of Music Technology.
2017	Invited speaker at The 5th Workshop on Fractional Calculus, Probability and Non-Local Operators: Applications and Recent Developments - Basque Center for Applied Mathematics.
2021	Remaking the world through Machine Learning Workshop: "Machine Learning in Healthcare"
2022	Remaking the world through Machine Learning Workshop: "Applied Mathematics as a Pre-processor for Data Driven Science in Medtech"
2022	UJ Public Lecture: "4IR in Medtech"

3.6 Referee Experience

3.6.1 Journal Refereeing

I have refereed articles for the following journals.

- Abstract and Applied Analysis
- British Journal of Mathematics & Computer Science
- Mathematical Problems in Engineering
- Journal of Basic and Applied Research
- Journal of Porous Media
- Journal of Scientific Research and Reports
- International Journal of Differential Equations
- Mathematical and Computational Applications
- Engineering Reports
- International Journal of Nonlinear Sciences and Numerical Simulation
- Journal of Computational and Theoretical Transport
- Mathematics and Computers in Simulation
- Journal of Computation Physics
- Fractal and Fractional
- East Asian Journal on Applied Mathematics
- Signal Processing
- Mathematics
- International Journal of Computer Mathematics
- Fractal and Fractional
- Applied Mathematics and Computation
- Chemical Engineering Communications
- Computers and Mathematics with Applications
- Journal of Nonlinear Science

3.6.2 Postgraduate Examination

2015 MSc in the School of Computer Science and Applied Mathematics, University of the Witwatersrand.

2016 MSc in the School of Mathematics, Statistics and Computer Science, University of Kwazulu-Natal.

2017 MSc in the School of Computer Science and Applied Mathematics, University of the Witwatersrand.

2018 MSc in the Department of Mathematics and Applied Mathematics, Univeristy of Pretoria.

2019 PhD in the Department of Mathematics and Applied Mathematics, Univeristy of Pretoria.

2022 MSc in the Department of Statistical Sciences University of Cape Town

3.6.3 Course External Examination

2020 External examiner for WTW386 in the Department of Mathematics and Applied Mathematics, Univeristy of Pretoria.

2021 External examiner for WTW386 in the Department of Mathematics and Applied Mathematics, Univeristy of Pretoria.

2020 External examiner for APPM1006 in the Department of Computer Science and Applied Mathematics, Univeristy of the Witwatersrand.

2021 External examiner for APPM1006 in the Department of Computer Science and Applied Mathematics, Univeristy of the Witwatersrand.

2022 - 2023 Various Undergraduate and Postgraduate courses in Applied Mathematics at Stellenbosch University and University of Pretoria.

3.6.4 NRF Rating Review

2015 Review of a researcher in South Africa (UFS) in the field of Fractional Differential Equations for NRF Rating.

2020 Review of NRF Thuthuka application.

2021 Review of a researcher in South Africa (UNISA) in the field of Symmetry Methods for Differential Equations for NRF Rating.

3.7 Professional Society Membership

2010 -	Member of The Golden Key Society International Honour Society.
2011 - 2015	Member of *Centre for Differential Equations, Continuum Mechanics and Applications.*
2014 -	Member of the South African Mathematical Society
2014 -	Member of the Society for Industrial and Applied Mathematics (SIAM).

4 Awards, Merits and Grants

2010	University of the Witwatersrand Postgraduate Merit Award
2010	Bronze Medal at the International Genetically Engineered Machine (IGEM) Competition hosted by the Massachusetts Institute of Technology, Boston, MA, USA.
2010	Rand Merchant Bank Gold Medal
2010	South African Mathematical Society Bronze Medal
2010	Starfield Prize for the Best Modelling Project in Computational and Applied Mathematics Honours
2011	University of the Witwatersrand Postgraduate Merit Award
2012	Awarded NRF Freestanding grant but declined due to employment opportunity within the School of Computational and Applied Mathematics.
2015	Award for best PhD in the School of Computational and Applied Mathematics.
2015 - 2017	NRF Thuthuka Grant.
2018 - 2021	NRF Competitive support for Unrated Researchers Grant.
2021	NRF Rating Incentive Funding.

4.1 Grantholder Funding Received

Year	Source	Amount
2015	NRF Thuthuka Grant 94005	R189,738.00
2016	NRF Thuthuka Grant 94005	R223,972.00
2016	Centre of Excellence in Mathematical and Statistical Sciences	R30,000.00
2017	NRF Thuthuka Grant 94005	R356,912.00
2018[1]	Blue Skies Research 104829	MSc Student Funding (OPEX funding for project R2,898,000.00)
2019	NRF Competitive support for Unrated Researchers Grant	R255,906.00
2020[2]	NRF Competitive support for Unrated Researchers Grant	R253,769.00
2021[2]	NRF Competitive support for Unrated Researchers Grant	R186,072.00
2021	NRF Rating Incentive Funding	R50,000.00
2025	Gates Foundation Funding (Strengthening Health and Disease Modelling for Public Health Decision Making in Africa.) for AI in Histpathology	R3,000,000.00

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5 Research

5.1 Publication in International Refereed Journals

1. **B. A. Jacobs** and E. Momoniat. *A novel approach to text binarization via a diffusion-based model.* **Applied Mathematics and Computation** 225 (2013): 446-460; `http://dx.doi.org/10.1016/j.amc.2013.09.048`.

2. **B. A. Jacobs** and C. Harley. *Two Hybrid Methods for Solving Two-Dimensional Linear Time-Fractional Partial Differential Equations.* **Abstract and Applied Analy-**

[1] funds earmarked for the remainder of the grant term.
[2] principal investigator on the grant.

sis (2014). http://dx.doi.org/10.1155/2014/757204. Hindawi Publishing Corporation, 2014.

3. **B. A. Jacobs** and E. Momoniat. *A Locally Adaptive, Diffusion Based Text Binarization Technique.* **Applied Mathematics and Computation** 269 (2015): 464-472; http://dx.doi.org/10.1016/j.amc.2015.07.091.

4. **B. A. Jacobs**. *A New Grünwald-Letnikov Derivative Derived From a Second-Order Scheme.* **Abstract and Applied Analysis** (2015). http://dx.doi.org/10.1155/2015/952057. Hindawi Publishing Corporation, 2015.

5. C. N. Angstmann, I. C. Donnelly, B. I. Henry, **B. A. Jacobs**, T. A. M. Langlands and J. A. Nichols. *From stochastic processes to numerical methods: A new scheme for solving reaction subdiffusion fractional partial differential equations.* **Journal of Computational Physics.** 307 (2016): 508-534. http://dx.doi.org/10.1016/j.jcp.2015.11.053

6. **B. A. Jacobs**. *High-order Compact Finite Difference and Laplace Transform Method for the Solution of Time-Fractional Heat Equations with Dirchlet and Neumann Boundary Conditions.* **Numerical Methods for Partial Differential Equations.** 32 (4) (2016): 1184-1199. http://dx.doi.org/10.1002/num.22046

7. C. N. Angstmann, B. I. Henry, **B. A. Jacobs** and A. V. McGann. *Integrablization of time fractional PDEs.* **Computers & Mathematics with Applications.** 73 (6) (2017): 1053-1062. https://doi.org/10.1016/j.camwa.2016.12.010

8. O. F. Egbelowo, C. Harley and **B. A. Jacobs**. *Nonstandard Finite Difference Method Applied to a Linear Pharmacokinetics Model.* **Bioengineering.** 4 (2) (2017): 40-61. http://dx.doi.org/10.3390/bioengineering4020040

9. C. N. Angstmann, B. I. Henry, **B. A. Jacobs** and A. V. McGann. *A time-fractional generalised advection equation from a stochastic process.* **Chaos, Solitons & Fractals.** (2017). https://doi.org/10.1016/j.chaos.2017.04.040

10. C. N. Angstmann, B. I. Henry, **B. A. Jacobs** and A. V. McGann. *Discretization of fractional differential equations by a piecewise constant approximation.* **Mathematical Modelling of Natural Phenomena.** 12 (2017): 23-36 https://doi.org/10.1051/mmnp/2017063.

11. C. N. Angstmann, B. I. Henry, **B. A. Jacobs** and A. V. McGann. *An explicit numerical scheme for solving fractional order compartment models from the master equations of*

a stochastic process. **Communications in Nonlinear Science and Numerical Simulation. 68 (2019): 188-202. IF: 3.181.**.
https://doi.org/10.1016/j.cnsns.2018.07.009

12. **B. A. Jacobs** and C. Harley. *Application of Hybrid Laplace Method to Nonlinear Time-Fractional Partial Differential Equations via Quasi-Linearization.* **Journal of Mathematics. IF: 0.68 (2018)** https://doi.org/10.1155/2018/8924547

13. C. N. Angstmann, B. I. Henry, **B. A. Jacobs** and A. V. McGann. *Numeric Solution of Advection-Diffusion Equations by a Discrete Time Random Walk Scheme.* **Numerical Methods for Partial Differential Equations. IF: 1.633** (2019) https://doi.org/10.1002/num.22448

14. M. Woolway, **B. A. Jacobs**, E. Momoniat, C. Harley. and D. Britz. *Convergence Analysis of the Frank-Kamenetskii Equation Using a High Order Shooting Method and Time-Marching Schemes.* **Entropy. IF: 2.419** 22(1) (2020): 84-101 https://doi.org/10.3390/e22010084

15. C. N. Angstmann, **B. A. Jacobs**, B. I. Henry, Zhuang Xu. *Intrinsic Discontinuities in Solutions of Evolution Equations Involving Fractional Caputo–Fabrizio and Atangana–Baleanu Operators.* **Mathematics. IF:1.747** 1(11) (2020): https://doi.org/10.3390/math8112023

16. **B. A. Jacobs**, and T. Celik. *Unsupervised document image binarization using a system of nonlinear partial differential equations.* **Applied Mathematics and Computation. IF: 4.091** 418 (2022): 126806. https://doi.org/10.1016/j.amc.2021.126806

17. N. Ndou, P. Dlamini, and **B. A. Jacobs**. *Enhanced Unconditionally Positive Finite Difference Method for Advection–Diffusion–Reaction Equations.* **Mathematics** 10.15 (2022): 2639. **IF: 2.592**

18. C. N. Angstmann, S. J. M. Burney, B. I. Henry, , **B. A. Jacobs** (2022). *Solutions of Initial Value Problems with Non-Singular, Caputo Type and Riemann-Liouville Type, Integro-Differential Operators.* **Fractal and Fractional**, 6(8), 436. **IF: 3.577**

19. S. Mtshali, **B. A. Jacobs** (2022). A Comparative Analysis of Physiologically Based Pharmacokinetic Models for Human Immunodeficiency Virus and Tuberculosis Infections. **Antimicrobial Agents and Chemotherapy**, 66(9), e00274-22. **IF: 5.191**

20. S. Mtshali, **B. A. Jacobs** (2023). On the Validation of a Fractional Order Model for Pharmacokinetics Using Clinical Data. Fractal and Fractional, 7(1), 84. **IF: 5.4**

21. **B. A. Jacobs**, F. Laurén, J. Nordström (2023). On the order reduction of approximations of fractional derivatives: an explanation and a cure. **BIT Numerical Mathematics**, 63(1), 17. **IF: 1.663**

22. C. N. Angstmann, S. J. M. Burney, B. I. Henry, **B. A. Jacobs**, Z. Xu. A Systematic Approach to Delay Functions. **Mathematics** 11.21 (2023): 4526. **IF: 2.4**

23. **B. A. Jacobs**, Image Trinarization Using a Partial Differential Equation: A Novel Approach to Automatic Sperm Image Analysis. **Applied Mathematical Modelling** 125 (2024): 704-727. **IF: 5.1**

24. N. Ndou, P, Dlamini, **B. A. Jacobs**. Solving the Advection Diffusion Reaction Equations by Using the Enhanced Higher-Order Unconditionally Positive Finite Difference Method. **Mathematics**, 12(7), 1009 (2024). **IF: 2.2**

25. N. Ndou, P, Dlamini, **B. A. Jacobs**. Developing higher-order unconditionally positive finite difference methods for the advection diffusion reaction equations. **Axioms**, 13(4), 247 (2024). **IF: 1.6**

26. **B. A. Jacobs**, I. Shaik, F. Lin. Predicting DNA fragmentation: A non-destructive analogue to chemical assays using machine learning. **arXiv** preprint arXiv:2409.13306 (2024).

27. Z. Xu, C. N. Angstmann, D. Han, B. I. Henry, S. J. M. Burney, **B. A. Jacobs**. An exact stochastic simulation method for fractional order compartment models. SIAM Journal on Applied **Mathematics**, 84(5), 2132-2151 (2024). **IF: 2.1**

28. S. Mtshali, **B. A. Jacobs**. Machine learning-based prediction of pharmacokinetic parameters for individualized drug dosage optimization. **International Journal of Information Technology**, 17(3), 1371-1385 (2025). **IF: 2.6**

29. K. Singh, **B. A. Jacobs**. A Network Based Model for Predicting Spatial Progression of Metastasis. **Bulletin of Mathematical Biology**, 87(5), 65. (2025) **IF: 2.2**

30. **B. A. Jacobs**, A. Morris, I. Shaik, F. Lin. Validation of an Artificial Intelligence Tool for the Detection of Sperm DNA Fragmentation Using the TUNEL In Situ Hybridization Assay. **arXiv** preprint arXiv:2510.11142 (2025).

5.2 Patents

1. I. Shaik, P. J. Lin, **B.A. Jacobs**. Visual analysis of sperm DNA fragmentation. US12406368B2 Vitruvianmd Pte Ltd. US Patent App. 19/289,222

5.3 Conference Proceedings

1. **B. A. Jacobs** and C. Harley. *A Comparison of Two Hybrid Methods for Applying the Time-Fractional Heat Equation to a Two Dimensional Function.* **AIP Conference Proceedings** 1558, 2119 (2013); `http://dx.doi.org/10.1063/1.4825955`.
2. **B. A. Jacobs** *A Differential Equation Based Approach to Sound Synthesis and Sequencing.* **International Computer Music Conference** Ann Arbor, MI: Michigan Publishing, University of Michigan Library. September 2016. `https://quod.lib.umich.edu/i/icmc/bbp2372.2016.109/1`
3. O. F. Egbelowo, C. Harley and **B. A. Jacobs** *Linear and nonlinear one compartment pharmacokinetic model: Nonstandard finite difference approach.* **Biomath Communications Supplement** 4, 1 (2016)
4. K. Prag, , M. Woolway, and **B. A. Jacobs**. *Optimising the Vehicle Routing Problem with Time Windows under Standardised Metrics.* 2019 6th International Conference on Soft Computing & Machine Intelligence (ISCMI). IEEE, (2019).

5.4 Invited Talks, Research Visits and Collaborations

2015 - 2019	Co-investigator with Prof Chrissie Rey in the School of Molecular and Cell Biology on NRF Blue Skies grant funded project.
April 2015 - July 2015	Research visit with Prof Bruce Henry at the University of New South Wales.
November 2016	Invited research visit to University of New South Wales, funded by UNSW, to conduct research with Dr. Chris Angstmann.
March 2016	Invited research visit by Dr. Chris Angstmann from the University of New South Wales to the University of the Witwatersrand.
November 2017	Invited speaker at The 5th Workshop on Fractional Calculus, Probability and Non-Local Operators: Applications and Recent Developments - Basque Center for Applied Mathematics.
November 2017	Invited research visit to Basque Center for Applied Mathematics (BCAM), Spain, funded by BCAM.
March 2017	Invited research visit by Dr. Chris Angstmann from the University of New South Wales to the University of the Witwatersrand.
February 2019	Invited research visit to Prof Tim Gebbie at the University of Cape Town.
March 2019	Invited research visit by Dr. Chris Angstmann from the University of New South Wales to the University of the Witwatersrand.
September 2019	Invited research visit to Prof Jan Nordstrom at Linkoping University, Sweden.
September 2023	Invited research visit to Prof Jan Nordstrom at Linkoping University, Sweden.

6 Supervision and Teaching

6.1 Honours Research Project Supervision

Continuous supervision of several students each year since 2012.

6.2 Postgraduate Supervision

2013	M. J. Woolway *Compressed Sensing: An Early Review* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 50% Co-supervision with Prof Momoniat.
2014 - 2015	L. Rogers *Finding The Sweet-Spot Of A Cricket Bat Using A Mathematical Approach* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 50% Co-supervision with Dr. Herbst.
2014 - 2017	T. Mothlele *Image Processing of Hyperspectral Images.* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 20% Co-supervision with Prof Sears
2015 - 2017	J. Shipton *Applications of Partial Differential Equations in Audio Signal Processing* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 100% supervision
2015 -	S. Maseko *Numerical Analysis and Simulations of Chemotaxis and Fractional Chemotaxis* for the degree of **MSc** in and Applied Mathematics. In Progress. 50% Co-supervision with Dr. Herbst. (Part-time)
2016 - 2017	D. Anderson *Machine Learning Techniques for Audio Signal Processing* for the degree of **MSc** in and Applied Mathematics. Completed. 50% Co-supervision with Dr. Klein.
2017	L. Borole *Numerical Investigation into Sound Synthesis through Differential Equations* for the degree of **MSc** in and Applied Mathematics. Completed. 100% supervision.
2018 - 2020	S. Schlebusch for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 100% supervision.
2018	K. Prag for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 50% Co-supervision with Dr. Woolway.
2015 - 2018	O. F. Egbelowo *Non-Standard Finite Difference Methods for the Pharmacokinetic Equations* for the degree of **PhD** in Computer Science and Applied Mathematics. Completed. 50 % Co-supervision with Prof Harley.
2018 - 2020	E. Conradie for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 100% supervision.
2018	S. Mtshali *The analysis of a physiologically based pharmacokinetic model subject to chronic infection* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed with Distinction. 100% supervision.
2018	D. Blankensee *Automatic mixing of musical compositions using machine learning* for the degree of **MSc** in Computer Science and Applied Mathematics. Completed. 100% supervision.
2018 - 2022	K. Singh for the degree of **MSc** in Computer Science and Applied Mathematics. Completed with Distinction. 100% supervision.

2020 - 2023 N. Ndou for the degree of **PhD** in Department of Mathematics and Applied Mathematics, UJ. Completed. 50% supervision.

2021 - S. Rametse for the degree of **PhD** in Department of Mathematics and Applied Mathematics, UJ. In Progress. 50% supervision.

2020 - 2026 S. Mtshali for the degree of **PhD** in Department of Mathematics and Applied Mathematics, UJ. Completed. 100% supervision.

7 Courses Taught

- Mathematical Modelling (APPM2007) (Wits), a second year level course taught to students majoring in Computational and Applied Mathematics for 100 - 150 students. 2011 - 2013

- Numerical Techniques (APPMC01) (Wits), a third year level course taught to students in Metallurgy and Industrial Engineering for 80 - 120 students. 2012

- Probabilistic Systems Analysis (ELEN3007) (Wits), a third year level course taught to students in Electrical and Information Engineering for 100 - 150 students. 2011 - 2017

- Numerical Methods (APPM4021) (Wits), an Honours level course taught to students in Advanced Mathematics of Finance for 10-15 students. 2013 - 2015

- Mathematical Modelling (APPM1006) (Wits), a first year level course taught to students majoring in Computational and Applied Mathematics for 350 - 420 students. 2015 - 2017

- Numerical Methods (APPM3017) (Wits), a third year level course taught to students majoring in Computational and Applied Mathematics for 50 - 100 students. 2018, 2020

- Numerical Methods (APPM2007) (Wits), a second year level course taught to students majoring in Computational and Applied Mathematics for 150 - 200 students. 2020

- Introduction to Statics (APM01A1) (UJ), a first year level course taught to students majoring in Applied Mathematics and Engineering for 300 - 600 students. 2021 - 2023

- Numerical Analysis B (APM8X13) (UJ), an honours level course taught to students majoring in Applied Mathematics 15 students. 2021 - 2022

- Lie Groups (APM8X20) (UJ), an honours level course taught to students majoring in Applied Mathematics 15 students. 2023 -

7.1 Teaching Development

2011	Attended and presented at the ‘Professionalisation of Teaching’ Symposium 2011.
2011 -	Conducted Lecturer Evaluations for courses taught from 2011.
2013	Peer evaluation conducted by Dr. Ann Cameron for Probabilistic Systems Analysis.
2014	Successful curriculum development for the inclusion of Stochastic Differential Equation into the second year Computational and Applied Mathematics curriculum.
2016	Successful integration of Introduction to Research Methods into the Computational and Applied Mathematics Honours programme.
2018	Science Teaching and Learning Centre Peer review conducted Prof Ann Cameron in 2018.
2018	Organisation of invited talk by Dr. Elizabeth Angstmann from the University of New South Wales, Australia entitled “Using technology to improve a large first year physics courses”.
2020 -	Innovation of cloud based programming instruction, together with autograded lab work, providing students immediate and detailed feedback for resubmission and multiple attempts, aiding learning and increased engagement with learning to code in python.

8 Conferences

- Presented *Noise reduction using the Fitzhugh-Nagumo equation*, at the joint international congress of the American and South African Mathematical Societies in Port Elizabeth, South Africa, November 2011.

- Presented *On the Application of Fractional Diffusion Equations with a Nonlinear Source Term to Image Processing* at the South African Symposium on Numerical and Applied Mathematics in Johannesburg, South Africa 2012.

- Presented *A Novel Approach to Text Binarization via a Diffusion-Based Model* at the South African Symposium on Numerical and Applied Mathematics in Stellenbosch, South Africa 2013.

- Presented *A Comparison of Two Hybrid Methods for Applying the Time-Fractional Heat Equation to a Two Dimensional Function.* at the 11th International Conference of Numerical Analysis and Applied Mathematics in Rodos, Greece, 2013.

- Presented *A New Grünwald-Letnikov Derivative Derived From a Second-Order Scheme* at the South African Symposium on Numerical and Applied Mathematics in Johannesburg, South Africa 2014.

- Presented a poster titled *High-order Compact Finite Difference and Laplace Transform Method for the Solution of the Time-Fractional Fokker-Planck Equation* at the East Asian Numerical Astrophysics Meeting at Kyung Hee University, Suwon, South Korea, 2014.

- Presented *Inverse Heat Conduction Problem in the Mixed Derivative Heat Equation* at the South African Symposium on Numerical and Applied Mathematics in Pretoria, South Africa 2015.

- Presented *A new scheme for solving some subdiffusion fractional partial differential equations*, at the South African Mathematical Societies in Johannesburg, South Africa, November 2015.

- Presented *Unsupervised Change Detection using a System of Nonlinear Partial Differential Equations* at the South African Symposium on Numerical and Applied Mathematics in Stellenbosch, South Africa 2016.

- Presented *A Differential Equation Based Approach to Sound Synthesis and Sequencing* at the International Computer Music Conference in Utrecht, Netherlands 2016.